FOR OFFICIAL USE

H

National
Qualifications
2016

Mark

X740/76/01

Human Biology
Section 1 — Answer Grid and Section 2

MONDAY, 9 MAY

1:00 PM – 3:30 PM

* X 7 4 0 7 6 0 1 *

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day Month Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *Page 02*.

SECTION 2 — 80 marks

Attempt ALL questions.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

* X 7 4 0 7 6 0 1 0 1 *

SECTION 1 — 20 marks

The questions for Section 1 are contained in the question paper X740/76/02.

Read these and record your answers on the answer grid on *Page 03* opposite.

Use **blue** or **black** ink. Do NOT use gel pens or pencil.

1. The answer to each question is **either** A, B, C or D. Decide what your answer is, then fill in the appropriate bubble (see sample question below).

2. There is **only one correct** answer to each question.

3. Any rough working should be done on the additional space for answers and rough work at the end of this booklet.

Sample Question

The digestive enzyme pepsin is most active in the

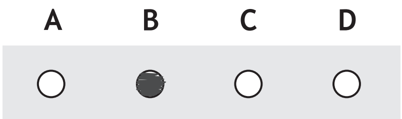
A mouth

B stomach

C duodenum

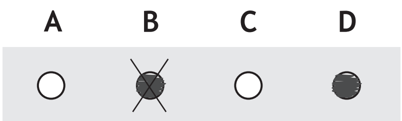
D pancreas.

The correct answer is **B** — stomach. The answer **B** bubble has been clearly filled in (see below).

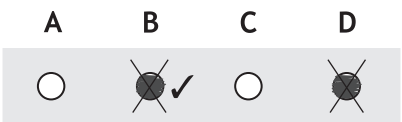


Changing an answer

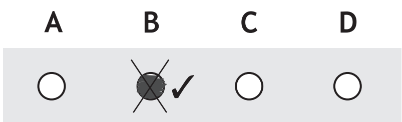
If you decide to change your answer, cancel your first answer by putting a cross through it (see below) and fill in the answer you want. The answer below has been changed to **D**.



If you then decide to change back to an answer you have already scored out, put a tick (✓) to the **right** of the answer you want, as shown below:



or



SECTION 1 — Answer Grid

OBJ20AD1

	A	B	C	D
1	○	○	○	○
2	○	○	○	○
3	○	○	○	○
4	○	○	○	○
5	○	○	○	○
6	○	○	○	○
7	○	○	○	○
8	○	○	○	○
9	○	○	○	○
10	○	○	○	○
11	○	○	○	○
12	○	○	○	○
13	○	○	○	○
14	○	○	○	○
15	○	○	○	○
16	○	○	○	○
17	○	○	○	○
18	○	○	○	○
19	○	○	○	○
20	○	○	○	○

X740760103

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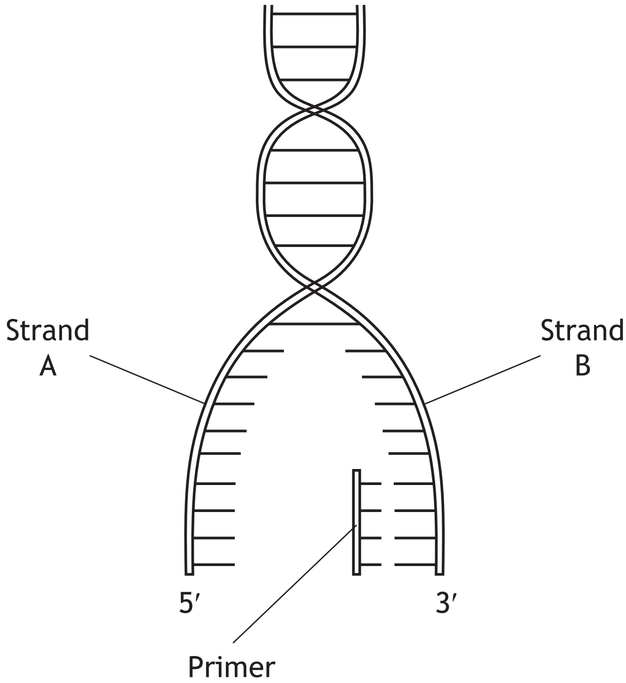
SECTION 2 — 80 marks

Attempt ALL questions

Note that Question 13 contains a choice.

1. The diagram below represents a stage in the process of DNA replication.



(a) (i) Name the type of bond which links the primer to strand B. **1**

____

(ii) Name the chemical group found at the 5′ end of a DNA strand. **1**

____

(b) Strand B is replicated continuously while strand A can only be replicated in fragments.

Explain why the strands are replicated in different ways. **1**

____

____

____

1. **(continued)**

(c) Describe the role of the following enzymes in DNA replication. **2**

DNA polymerase ____________________

____

Ligase ____________________

____

MARKS | DO NOT WRITE IN THIS MARGIN

2. At the start of polypeptide synthesis in a cell, DNA transcription occurs in the nucleus to form mRNA.

The sequence of bases from a section of a DNA strand is shown below.

...... C A C G A T C G A T A G G A T ......

(a) (i) State the sequence of bases in the primary mRNA transcript formed from this strand of DNA. 1

(ii) State the term used to describe the coding regions of a primary mRNA transcript. 1

(iii) Name the process by which the coding regions of a primary mRNA transcript are joined together to produce a mature mRNA transcript. 1

(iv) The sequence of bases in the mature mRNA transcript, formed from the section of the DNA strand, is shown below.

...... G U G C U A U C C U A ......

Using this mature mRNA transcript, state the order of bases in the intron present in the primary mRNA transcript. 1

(b) State the location for the translation of a mature mRNA transcript into a polypeptide. 1

(c) Describe **one** form of post-translational modification of a polypeptide. 1

[Turn over

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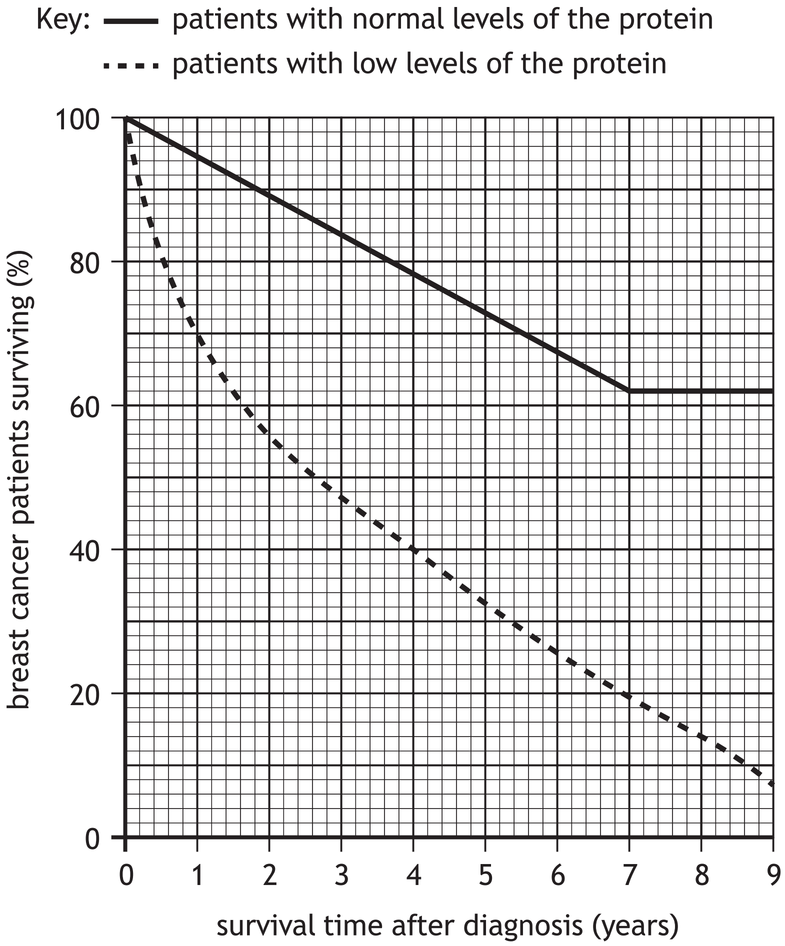
MARKS DO NOT WRITE IN THIS MARGIN

3. A naturally occurring cell protein (nm23) has been shown to inhibit the activity of cancer cells.

Individuals produce varying levels of this protein depending on their genetic make-up.

The graph below shows the results of a 9 year study of women diagnosed with breast cancer. The women were divided into two groups according to their production of the protein.



(a) (i) In a city, 1000 women were diagnosed with breast cancer.

Of these women, 900 had normal levels of the protein while 100 had low levels.

Using the results from the study, calculate how many of the 1000 women would be expected to survive for 4 years after diagnosis. **1**

Space for calculation

3. (a) (continued)

(ii) **Use data from the graph** to describe the changes in the percentage of surviving breast cancer patients with normal levels of the protein during the study. **2**

(b) Describe how cancer can develop and spread through the body. **3**

[Turn over

4. A student carried out an investigation into the effect of physical activity on respiration rate.

The rate of respiration of six individuals was measured after carrying out three different activities for five minutes.

Immediately after completing the activity, each individual breathed into a bottle containing a pH indicator solution. This indicator changes colour from blue to yellow in the presence of a high concentration of carbon dioxide.

Figure 1 – Apparatus used

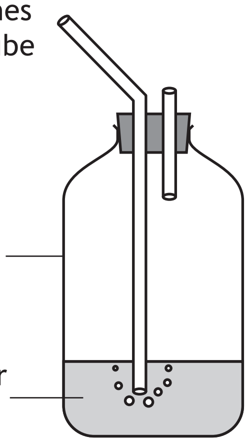


Table 1 – Results of Investigation

Individual	*Time taken for pH indicator to turn yellow* (s)		
	Activity 1 resting	Activity 2 walking	Activity 3 running
1	33	28	22
2	29	14	11
3	22	16	12
4	30	26	20
5	44	35	22
6		31	21
Average time taken (s)	33	25	18

(a) State **two** variables which would have to be kept constant when setting up the apparatus shown in **Figure 1**. **2**

1 ____

2 ____

(b) Calculate the time taken for the indicator to turn yellow after individual 6 had completed Activity 1. **1**

Space for calculation

______________ s

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4. (continued)

(c) Describe how the student increased the reliability of the results. **1**

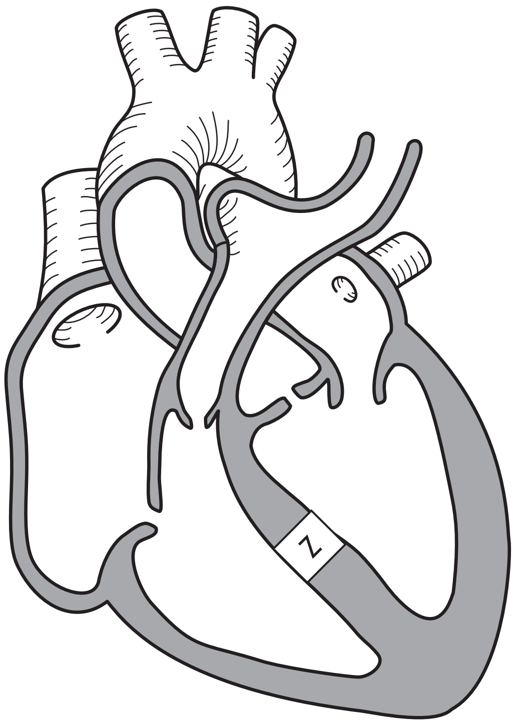
(d) Construct a bar graph to show the average results obtained in this investigation. **2**

(Additional graph paper, if required, can be found on *Page 31*)

(e) State a conclusion that can be drawn from the results of this investigation. **1**

(f) Suggest an explanation for the results obtained in this investigation. **1**

5. The diagram below represents the structure of the heart and its associated blood vessels.



(a) On the diagram, **label** the pulmonary artery with the letter P. **1**

(b) Sometimes babies can be born with a ventricular septal defect (VSD) in which a "hole" occurs at point Z in the heart.

Explain how the presence of this hole would affect the oxygen concentration of the blood leaving the heart through the aorta. **2**

5. (continued)

(c) Babies with a VSD sometimes have irregular heart rhythms. This can be detected by recording the electrical activity from the heart.

(i) Name the chamber of the heart in which this electrical activity originates. **1**

____

(ii) Name the type of graph that displays such patterns of electrical activity. **1**

____

(d) Babies with a VSD often have a lower stroke volume than babies who have a normal heart structure.

Despite this, both groups of babies often have similar cardiac outputs.

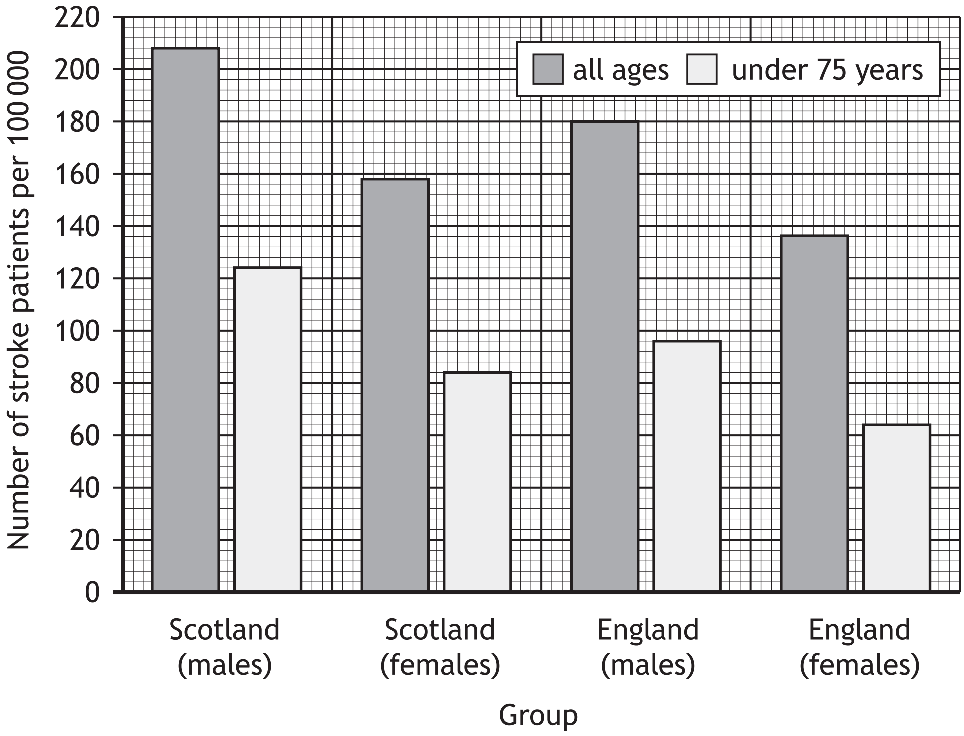
Suggest how babies with a VSD are able to achieve a similar cardiac output to babies with a normal heart structure. **1**

____

____

[Turn over

6. The graph below shows the number of stroke patients in different groups from Scotland and England in 2007.



(a) (i) Calculate the difference in the number of male stroke patients of all ages in Scotland and England in 2007. **1**

Space for calculation

________________ per 100 000

(ii) Explain the importance of presenting the data as the number of stroke patients **per 100 000**. **1**

____

____

6. (a) (continued)

(iii) Scotland's population was 5·1 million in 2007.

Calculate the number of female stroke patients in Scotland under 75 years of age in this year. **1**

Space for calculation

____

(iv) Express, as a simple whole number ratio, the number of male stroke patients under 75 years of age compared to female stroke patients under 75 years of age in England in 2007. **1**

Space for calculation

__________ : __________

male patients female patients

(b) Describe what causes a stroke. **1**

____

____

(c) Paralysis occurs when voluntary muscle is unable to contract.

Explain how a stroke could lead to muscle paralysis on the left side of the body. **2**

____

____

____

____

____

MARKS | DO NOT WRITE IN THIS MARGIN

7. The table below contains information about five obese patients who attended a weight loss clinic for 12 weeks.

Patient	*Height* (m)	*Starting weight* (kg)	*Starting BMI*	*Final Weight* (kg)	*Final BMI*
P	1·74	92	30·5	82	27·2
Q	1·68	98	34·8	90	32·1
R	1·81	104	31·8	97	29·7
S	1·89	121	33·9	113	31·4
T	1·90	100	32·3	94	

(a) (i) Calculate the final BMI of patient T. **1**

Space for calculation

Final BMI = ____________________

(ii) State why patient Q was still classed as obese after 12 weeks. **1**

____

____

(b) Explain why all the patients were advised to exercise regularly to increase their weight loss. **1**

____

____

____

* X 7 4 0 7 6 0 1 1 6 *

7. (continued)

(c) Rugby players may have a BMI which indicates that they are obese.

Suggest why a BMI reading may not be a reliable indicator of obesity in rugby players. **1**

[Turn over

8. Statins are drugs which reduce the production of cholesterol in the liver. A year-long trial was carried out to investigate the effects of taking a newly-developed statin on blood cholesterol levels.

Sixty individuals with raised blood cholesterol levels were selected and divided into two groups of thirty.

Individuals in Group 1 were prescribed a capsule, containing 20 mg of the statin, to take each day. Individuals in Group 2 were the control group. At two-monthly intervals, blood samples were taken from all individuals and their blood cholesterol levels measured.

The results are shown in the table below.

	Average blood cholesterol level (mmol/l)	
Month of trial	Group 1	Group 2
0	6·3	6·3
2	6·3	6·3
4	6·3	6·1
6	6·3	6·3
8	5·6	6·1
10	5·3	6·2
12	5·1	6·1

(a) **Using the results in the table**, give **one** reason why this drug might be **2**

recommended __________________________________

not recommended __________________________________

(b) Suggest what was prescribed to the individuals in Group 2 during this trial. **1**

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8. (continued)

(c) Describe the design features which would have been used to ensure that this was both a randomised and a double-blind trial. **2**

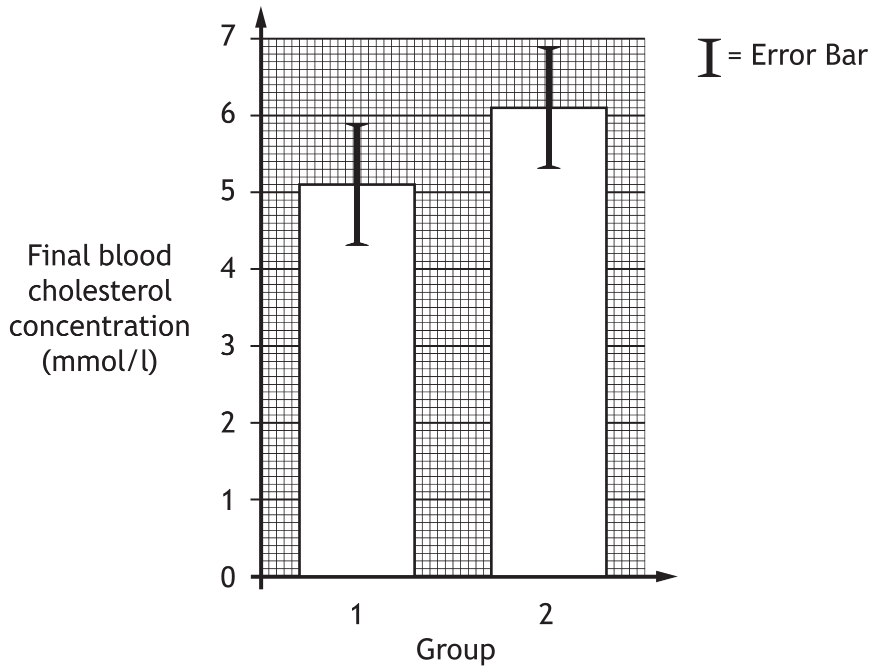
randomised ______________________________

____

double-blind ______________________________

____

(d) The bar graph below summarises the data collected in the final month of this trial.



Using evidence from the bar graph, suggest why it was decided that this statin was not worth further development. **1**

____

____

(e) Describe **one** function of cholesterol in the human body. **1**

____

____

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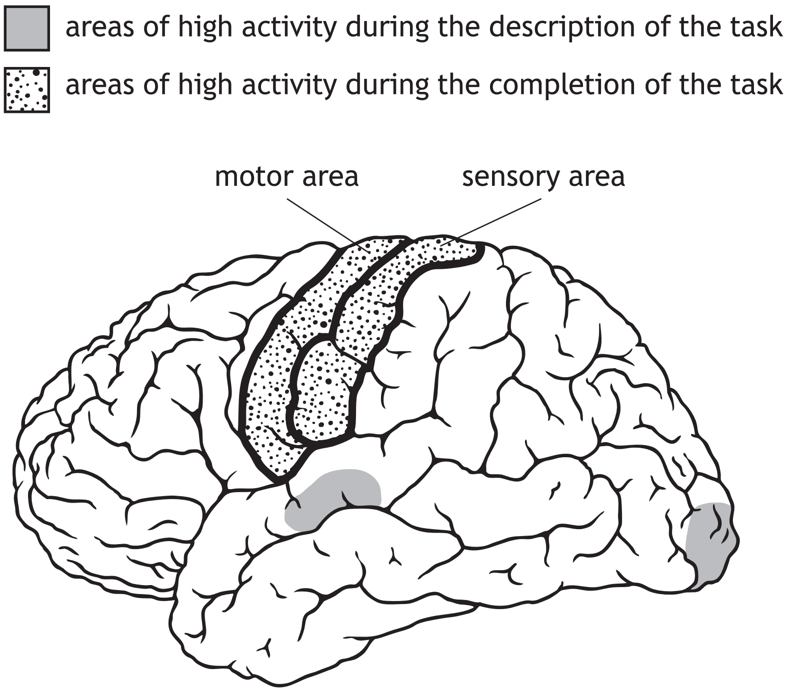
9. The diagram below represents areas of high activity in a part of the brain of an individual as a task is described to them which they then complete.



(a) Name the part of the brain shown in the diagram. **1**

____

(b) Explain how the diagram supports the suggestion that there is localisation of function in the brain. **1**

____

____

(c) Explain the high level of brain activity during the description of the task. **1**

____

____

9. (continued)

(d) The task was to fold a piece of paper.

Explain why the diagram shows high levels of activity in the sensory and motor areas. **2**

Sensory area ______________________________

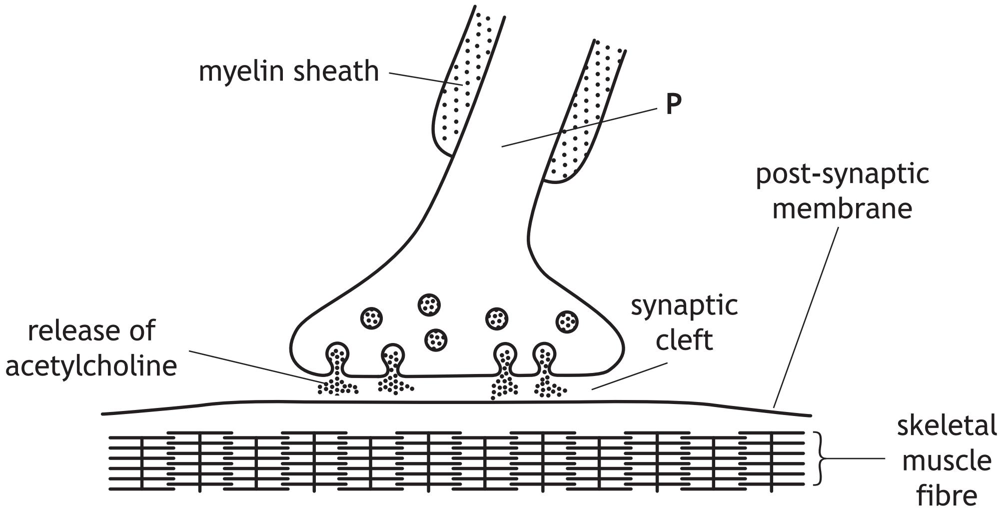
____

Motor area ______________________________

____

[Turn over

10. The diagram below shows a synapse in skeletal muscle of a weightlifter.



(a) Name the part of the motor neuron labelled **P**. **1**

____

(b) Describe what acetylcholine does when it reaches the post-synaptic membrane. **1**

____

____

(c) Name the type of skeletal muscle fibres which will be the most common in the arm muscles of a champion weightlifter. **1**

____

MARKS | DO NOT WRITE IN THIS MARGIN

10. (continued)

(d) Nicotine is a drug that is an agonist of acetylcholine.

(i) Explain how an agonist works. **1**

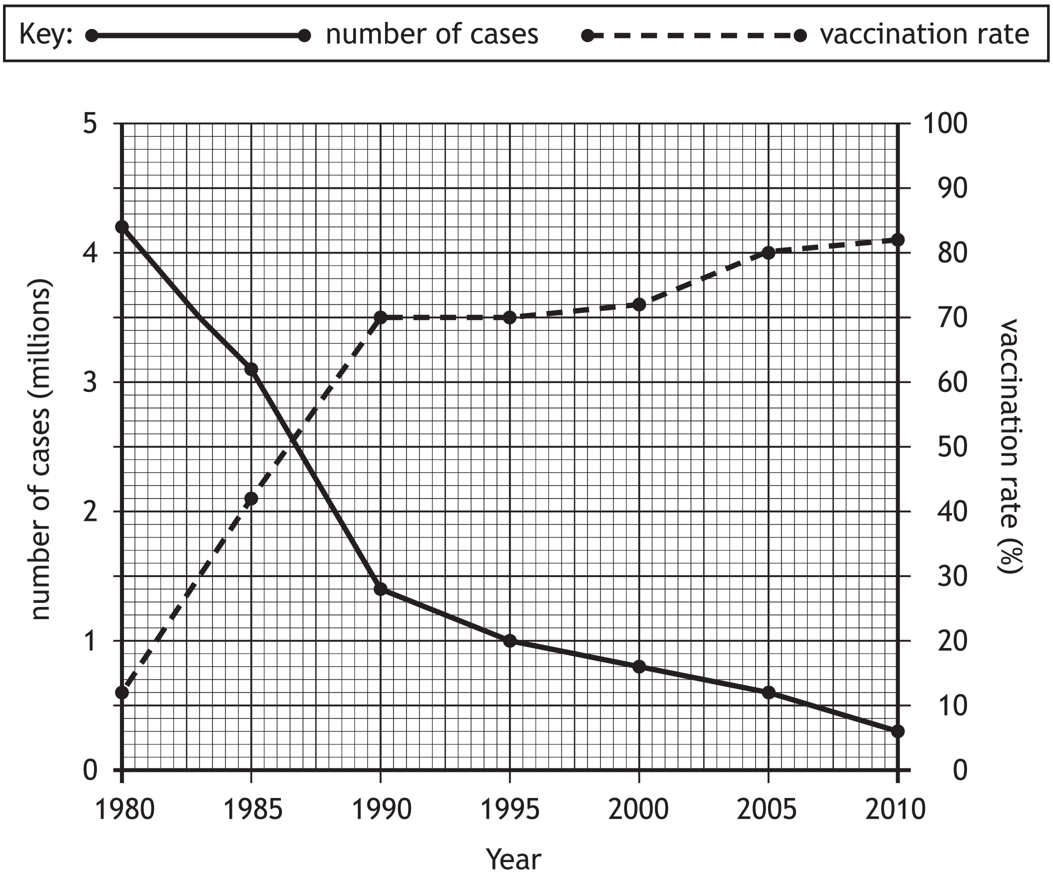
(ii) Suggest how nicotine induces feelings of pleasure and so reinforces smoking behaviour. **1**

[Turn over

* X 7 4 0 7 6 0 1 2 3 *

MARKS | DO NOT WRITE IN THIS MARGIN

11. The graph below shows the number of cases of measles that occurred in the world between 1980 and 2010. It also shows the global vaccination rate against measles over the same period.



(a) State how many cases of measles there were in 1985. **1**

____

(b) State the vaccination rate when there were 3·5 million cases of measles in the world. **1**

________________________%

(c) Calculate the percentage decrease in the number of cases of measles between 1995 and 2010. **1**

Space for calculation

____

* X 7 4 0 7 6 0 1 2 4 *

11. (continued)

(d) In many countries herd immunity has been established against measles.

(i) Explain why people in these countries who have not been vaccinated are still protected against measles. **1**

(ii) Suggest **one** reason why widespread vaccination programmes against measles are not possible in all countries of the world. **1**

(e) In 2010 the population of the world was 6900 million.

Using the information from the graph, calculate how many people in the world had not been vaccinated against measles in 2010. **1**

Space for calculation

____________ million

(f) The World Health Organisation (WHO) has set a goal of eliminating measles worldwide by 2020.

Explain how the information in the graph indicates that this goal can be achieved. **1**

[Turn over

12. Specific cellular defences are part of the body's immune system and give protection against individual types of pathogen.

(a) (i) Explain how a clonal population of lymphocytes would be formed when a pathogen invades the body. **2**

(ii) Describe the role of phagocytes in the specific immune response. **2**

(b) Failure of the immune system can lead to conditions such as allergy and autoimmune disease.

Choose **one** of these conditions and complete the table below with information about it. **2**

Condition ____________________

Type of white blood cell involved	*Description of immune system failure*

MARKS DO NOT WRITE IN THIS MARGIN

13. Answer **either** A **or** B in the space below.

Labelled diagrams may be used where appropriate.

A Discuss the causes, development and associated health problems of atherosclerosis. **8**

OR

B Discuss the diagnosis, treatment and role of insulin in Type 1 and Type 2 diabetes. **8**

* X 7 4 0 7 6 0 1 2 7 *

MARKS | DO NOT WRITE IN THIS MARGIN

ADDITIONAL SPACE FOR ANSWER to Question 13

[END OF QUESTION PAPER]

* X 7 4 0 7 6 0 1 2 8 *

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

Additional graph paper for Question 4 (d)

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